

nf-cavalier: A Nextflow Pipeline for Rare Disease Variant Prioritization and Reporting

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Software

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Summary

nf-cavalier is a Nextflow pipeline that automates genomic variant annotation, filtering, and reporting for individuals with rare Mendelian diseases. The pipeline takes as input variant callsets for an individual, family, or rare disease cohort, together with a target gene panel or a phenotype of interest. Variants are then filtered using various customisable criteria, including predicted gene consequence, computational pathogenicity predictions, population frequency, and familial segregation. The sequencing data for candidate variants is then visualised for human review. Candidate variant results are returned in user-friendly output formats, including interactive HTML reports and PowerPoint slide decks, with embedded links to external resources that enable rapid review by clinical research teams. **nf-cavalier** is maintained on GitHub ([bahlolab/nf-cavalier](https://github.com/bahlolab/nf-cavalier)) and licensed under the permissive MIT open-source licence.

Statement of need

Identifying candidate pathogenic genetic variants from genome sequencing data is a complex task that requires coordinating a wide range of software and data sources. Millions of variant calls are generated per individual and must be interpreted alongside known disease gene associations (e.g., PanelApp ([Martin et al., 2019](#)), HPO ([Köhler et al., 2021](#))), gene and transcript models (e.g., GENCODE ([Frankish et al., 2021](#))), computational pathogenicity predictions (e.g., CADD ([Schubach et al., 2024](#))), population frequency data (e.g., gnomAD ([Chen et al., 2024](#))) and known pathogenic variant databases (e.g., ClinVar ([Landrum & Kattman, 2018](#))). Variant inheritance patterns must be considered in familial cases, with population frequency filters tailored to dominant, recessive, and compound heterozygous inheritance patterns. Finally, candidate variants must be reported, and the sequencing data supporting the variant visualised for human review prior to further validation, segregation, and pathogenicity classification.

This task is further complicated by the need to jointly interpret short variants (single-nucleotide variants (SNVs; 1 bp) and insertion-deletion variants (indels; 2 – 50 bp)) and structural variants (larger alterations; > 50 bp), which are typically identified using separate workflows.

Several software tools are commonly used for the individual tasks carried out in this workflow. For example, VEP ([McLaren et al., 2016](#)) handles variant annotations and consequence prediction based on specified gene models, but does not support inheritance-based filtering or variant visualisation. slivar ([Pedersen et al., 2021](#)) enables rapid querying of variant call format (VCF) files with an expressive syntax that allows specific inheritance models to be

interrogated; however, it does not perform annotation or visualisation. Tools such as igv-reports (Robinson et al., 2026) and SVPV (Munro et al., 2017) generate visualisations of sequencing data to aid variant review, but do not perform prioritisation. As such, there is a need for tools that orchestrate and coordinate a variety of task-specific applications into a unified variant prioritisation and reporting framework, improving both reproducibility and user-friendliness. Such integrated workflows are easier to maintain, apply consistently, and validate than fragmented, ad hoc approaches.

nf-cavalier automates this process in a flexible, extensible, and reproducible manner, generating user-friendly outputs in interactive HTML and PowerPoint formats that can be shared and reviewed by multiple collaborators. Metadata describing gene lists and runtime parameters are embedded in the HTML reports, enabling previous analyses to be audited and reproduced. By integrating annotation, inheritance-based filtering, visualisation, and reporting into a single workflow, **nf-cavalier** reduces the manual effort required for rare disease variant prioritisation while improving reproducibility. The target audience for **nf-cavalier** is clinical research teams exploring the genetic causes of rare diseases.

State of the field

There are several comparable tools for rare disease variant prioritisation; most notably seqr (Pais et al., 2022), Scout (Stranneheim et al., 2021), and MOLGENIS-VIP (Maassen et al., 2025) (see Table 1). Of these, VIP is the closest comparator to **nf-cavalier**, as both tools are Nextflow-based and produce static HTML reports without requiring a server or database backend. This makes them suitable for institutional HPC environments where persistent server infrastructure is typically unsupported or undesirable. In contrast, seqr and Scout depend on continuously running web services, which makes them out of reach for many research teams due to ongoing costs and infrastructure requirements. The recommended seqr deployment uses Kubernetes to orchestrate multiple interdependent services, including SQL databases (PostgreSQL, ClickHouse), Redis (cache), the seqr application, and a pipeline-runner service for loading variants into the databases. Scout is simpler, with the recommended deployment using two Docker services: MongoDB and the Scout web application. Many institutional HPC environments restrict or do not support Kubernetes and Docker deployments.

nf-cavalier and VIP are designed for different use cases. VIP targets production-oriented clinical or diagnostic variant classification, while **nf-cavalier** targets exploratory, tunable research candidate discovery and prioritisation. VIP encodes interpretation rules as a configurable decision tree in JSON, combining evidence criteria to assign variants to pathogenicity classifications based on ACMG guidelines. In contrast, **nf-cavalier** does not attempt automated classification; instead, it applies configurable filtering criteria to identify candidate variants for downstream review and interpretation. These criteria, including allele frequency, genotype quality, consequence class, and gene-list membership, are exposed as independently configurable, named parameters. For example, `FILTER_SHORT_POP_DOM_MAX_AF = 0.0001` sets the maximum gnomAD allele frequency allowed for dominantly inherited short variants. This provides a transparent representation of the filtering strategy, allowing researchers to inspect, understand, and iteratively refine individual filtering criteria in a straightforward manner.

Software design

Variant prioritisation pipelines are subject to substantial churn – new annotation sources become available, new pathogenicity-prediction methods are released, and new classes of variation are recognised. **nf-cavalier** is designed to allow new annotation datasets, pathogenicity predictions, and population frequency resources to be incorporated with minimal effort, and the framework is left open to incorporating new approaches as they are released. The pipeline is fundamentally anchored in the VCF specification, allowing compatibility with a range of

89 upstream variant callers and annotation resources. This makes **nf-cavalier** particularly suitable
90 for research pipelines, which are often at the cutting edge of new variant discovery.

Table 1: Comparison of **nf-cavalier** to leading variant prioritisation tools.

Tool	nf-cavalier	MOLGENIS-VIP	seqr	Scout
Publication year	2026	2025	2022	2021
Required Preprocessing	SV and SNV/Indel variant calling	None	GATK4 SNV/Indel calling	nf-core/raredisease variant calling
Dependencies	Nextflow and container engine	Nextflow and container engine	Kubernetes, Live Server	Docker, Live Server
Variant types	SNV/Indel, SV	SNV/Indel, SV, STR	SNV/indel, SV [†]	SNV/indel, SV, STR
QC	Somalier -relate and -ancestry, Sample contamination	None	None	None
Gene lists supported	PanelApp, HPO, genomic region, HGNC locus, Genes4Epilepsy [§] , custom	HPO [‡]	PanelApp, HPO, genomic region, custom	PanelApp, HPO, custom
Filtering	Nextflow parameters, R-script	JSON Decision tree, HTML GUI	Web GUI	Web GUI
Outputs	Static HTML, PowerPoint slides	Static HTML	Web server	Web server

91 [†] SV support in seqr is currently limited to in-house users at the Broad Institute, as the SV
92 calling and loading pipelines have not been made available at the time of writing. [‡] VIP does
93 not filter output based on gene panels, but can annotate HPO terms for user-level filtering. [§]
94 (Oliver et al., 2023)

95 The pipeline has been implemented in Nextflow, leveraging Nextflow's framework for
96 reproducibility, containerisation, and ability to run on local machines, high-performance
97 computing clusters, and cloud environments. The pipeline consists of three modules:
98 annotation, QC, and reporting (Figure 1). We also include a one-time setup pipeline that
99 automates the download of required reference datasets, as well as a script to download a
100 demonstration dataset based on the 1000 Genomes CEPH trio for testing and evaluation.

101 Annotation takes cohort VCFs (short and/or structural variants) as input, splits them into
102 shards for parallel processing, and performs variant normalisation and annotation with standard
103 tools bcftools (Danecek et al., 2021), vcfanno (Pedersen et al., 2016), VEP (McLaren et al.,
104 2016), and SVAfotate (Nicholas et al., 2022). Sharding improves scalability, reducing runtime
105 from days to hours depending on available compute resources. vcfanno is employed to rapidly
106 incorporate pre-calculated variant-specific metrics for short variants, including population allele
107 frequencies from gnomAD v4.1, known pathogenicity classifications from ClinVar, predicted
108 deleteriousness scores from CADD, as well as arbitrary user-specified annotation sources. VEP
109 is used to add gene-level consequence predictions as well as annotations that depend on
110 transcript models, including REVEL (Ioannidis et al., 2016), AlphaMissense (Cheng et al.,
111 2023), and SpliceAI (Jaganathan et al., 2019) scores. For structural variants, SVAfotate is
112 used to transfer gnomAD v4.1 SV frequencies using an overlap-based matching approach.

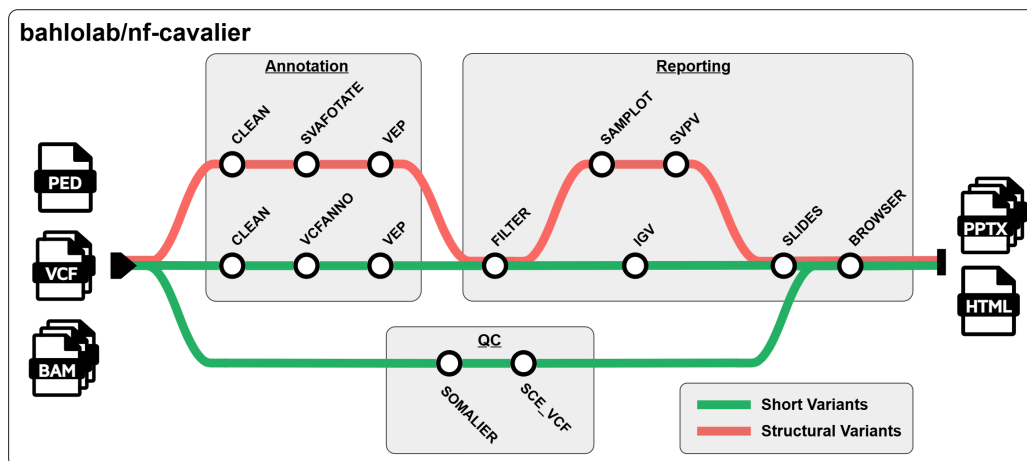


Figure 1: Schematic overview of nf-cavalier pipeline.

Filtering operates on a per-family basis and applies user-configurable filters for genes of interest (derived from PanelApp, HPO, genomic regions, or custom inputs), genic consequence (e.g. VEP impact or consequences), and deleteriousness predictions. By default, all reported ClinVar pathogenic/likely pathogenic variants in a gene of interest are retained, and all reported ClinVar benign or likely benign variants are excluded. Variants are then matched to inheritance models – all affected family members must carry dominant variants; recessive variants must be present only in a homozygous state in affected family members; and compound heterozygous pairs must be present together only in affected family members. Allele frequency filters are then applied depending on inheritance model, with the default dominant threshold set at 1/10,000 and the default recessive threshold set at 1/100.

The reporting stage visualises candidate variants using igv-reports, SVPV, or Samplot to support manual review and quality control. Reports are compiled into PowerPoint or PDF slide decks (Figure 2) and published alongside a static HTML report that enables interactive querying and exploration of variants (Figure 3). The HTML report also provides live access to other resources via public APIs, including PanelApp, HPO, ClinVar, gnomAD, and the UCSC genome browser.

In addition to the annotation, filtering, and reporting stages, the pipeline includes a QC workflow that runs somalier (Pedersen et al., 2020) and SCE-VCF (Lu et al., 2023) to assess relatedness, sex concordance, genetic ancestry, and contamination, enabling users to identify problematic samples and verify pedigree relationships.

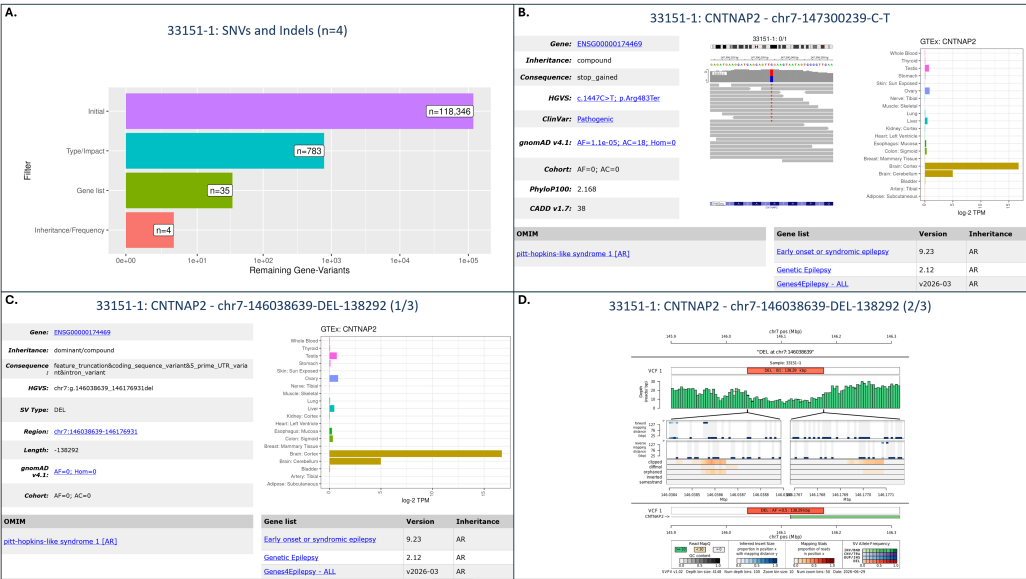


Figure 2: Example *nf-cavalier* variant PowerPoint output slides for a research participant with compound heterozygous pathogenic variants in the gene *CNTNAP2*. A) Summary slide detailing counts after each level of SNV/Indel filtering was applied, resulting in four candidate variants for this example analysis. B) PowerPoint output slide with details on a stop-gain variant in *CNTNAP2*, known to be pathogenic in ClinVar. Blue text represents hyperlinks to external resources. C) PowerPoint output slide detailing a 138 kb deletion affecting *CNTNAP2*, flagged by *nf-cavalier* as potentially compound heterozygous. D) *nf-cavalier* PowerPoint slide embedding an SVPV plot of the structural variant for inspection and QC. This case has been published in Munro et al. (Munro et al., 2026).

Research impact statement

nf-cavalier has been used in several studies investigating known and novel causes of rare disease in large-scale genome and exome sequencing cohorts, benefiting from extensive real-world application testing over five years. These include the Austin Health Adult Undiagnosed Disease Program (AHA-UDP), the first adult-focused undiagnosed disease program in Australia, which resulted in diagnoses for 16/50 (32%) participating families (Wallis et al., 2024). More recently, *nf-cavalier* has been used to search for molecular causes in a large cohort of individuals with developmental and epileptic encephalopathy (DEE), resulting in diagnoses for 37/242 (15%) of participants, including nine structural variants (Munro et al., 2026), and in the identification of a complex structural variant in *FBRSL1* associated with a severe DEE (Cohen-Vig et al., 2026). Furthermore, *nf-cavalier* was used to assist in the landmark discovery of an intronic repeat expansion in *FGF14* as the most common cause of late-onset cerebellar ataxia, where it was used to exclude alternative explanations for pathogenicity (Rafehi et al., 2023). The CAVALIER R package, a precursor to the current Nextflow implementation, was also used similarly to assist with the discovery of the RFC1 repeat expansion causing cerebellar ataxia with neuropathy and bilateral vestibular areflexia syndrome (CANVAS) (Rafehi et al., 2019).

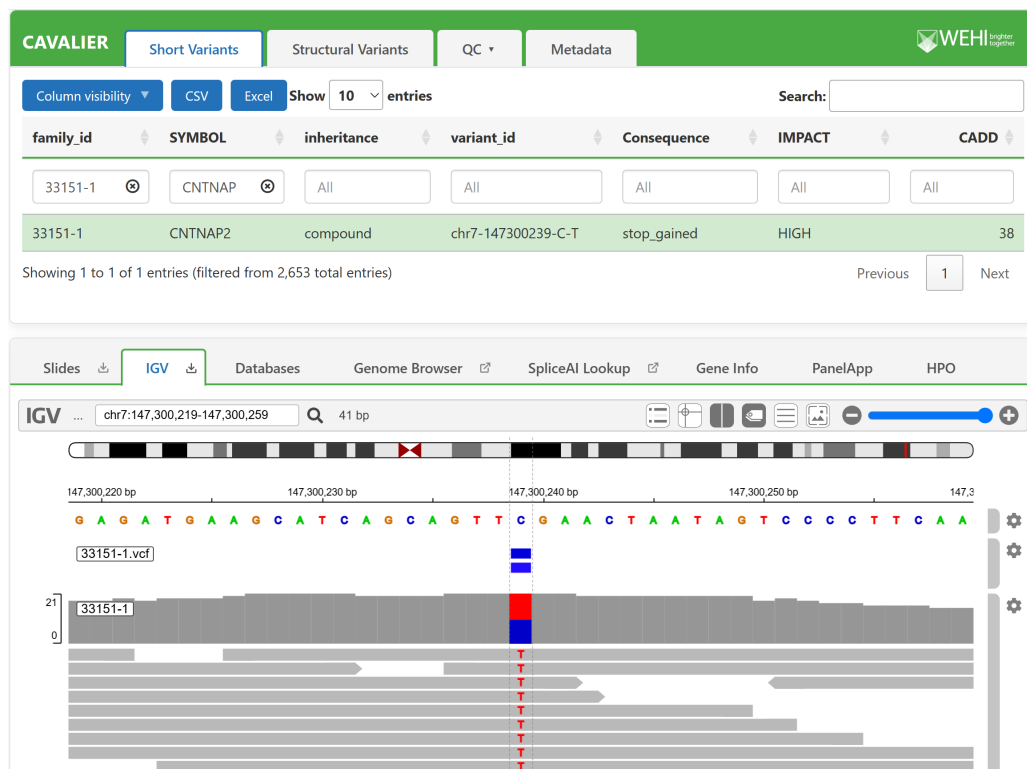


Figure 3: The CAVALIER interactive variant browser interface. The upper panel allows filtering of short and structural variants using customisable columns and includes tabs for QC data and run metadata. The lower panel enables exploration of variants and genes, with igv-reports (pictured) and other external APIs embedded.

149 AI usage disclosure

The majority of the **nf-cavalier** source code was written and tested without generative AI assistance. Since March 5, 2026 (GitHub commit 'a1196d9'), Claude Code has been used for coding and documentation assistance, using a mixture of Sonnet 4.6 and Opus 4.7-4.8 models depending on task complexity. AI assistance was most useful for creating the interactive HTML output, updating and expanding the documentation, and implementing the pipelines to download the reference data and the test dataset. The AI tools were given clear, specific instructions on implementation and design, and all AI-generated content has been reviewed and tested by the authors. The authors made key decisions and take full responsibility for the software's reliability and maintenance. The Claude web interface has also been used to assist with grammar, clarity, and phrasing for this manuscript.

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173 References

- 174 Chen, S., Francioli, L. C., Goodrich, J. K., Collins, R. L., Kanai, M., Wang, Q., Alföldi, J.,
 175 Watts, N. A., Vittal, C., Gauthier, L. D., Poterba, T., Wilson, M. W., Tarasova, Y., Phu,
 176 W., Grant, R., Yohannes, M. T., Koenig, Z., Farjoun, Y., Banks, E., ... Karczewski, K. J.
 177 (2024). A genomic mutational constraint map using variation in 76,156 human genomes.
 178 *Nature*, 625(7993), 92–100. <https://doi.org/10.1038/s41586-023-06045-0>
- 179 Cheng, J., Novati, G., Pan, J., Bycroft, C., Žemgulytė, A., Applebaum, T., Pritzel, A., Wong,
 180 L. H., Zielinski, M., Sargeant, T., Schneider, R. G., Senior, A. W., Jumper, J., Hassabis, D.,
 181 Kohli, P., & Avsec, Ž. (2023). Accurate proteome-wide missense variant effect prediction
 182 with AlphaMissense. *Science*, 381(6664), eadg7492. <https://doi.org/10.1126/science.adg7492>
- 183
- 184 Cohen-Vig, L., Munro, J. E., Reid, J., Witkowski, T., Sikta, N., Kraus, D., Bennett, M.
 185 F., Scheffer, I. E., Hildebrand, M. S., Bahlo, M., & Berkovic, S. F. (2026). A complex
 186 duplication overlapping FBRSL1 implicated in a developmental and epileptic encephalopathy.
 187 *medRxiv*. <https://doi.org/10.64898/2026.03.30.26349353>
- 188 Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A.,
 189 Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools
 190 and BCFtools. *Gigascience*, 10(2). <https://doi.org/10.1093/gigascience/giab008>
- 191 Frankish, A., Diekhans, M., Jungreis, I., Lagarde, J., Loveland, J. E., Mudge, J. M., Sisu,
 192 C., Wright, J. C., Armstrong, J., Barnes, I., Berry, A., Bignell, A., Boix, C., Carbonell
 193 Sala, S., Cunningham, F., Di Domenico, T., Donaldson, S., Fiddes, I. T., García Girón,
 194 C., ... Flicek, P. (2021). GENCODE 2021. *Nucleic Acids Res.*, 49(D1), D916–D923.
 195 <https://doi.org/10.1093/nar/gkaa1087>
- 196 Ioannidis, N. M., Rothstein, J. H., Pejaver, V., Middha, S., McDonnell, S. K., Baheti,
 197 S., Musolf, A., Li, Q., Holzinger, E., Karyadi, D., Cannon-Albright, L. A., Teerlink, C.
 198 C., Stanford, J. L., Isaacs, W. B., Xu, J., Cooney, K. A., Lange, E. M., Schleutker,
 199 J., Carpten, J. D., ... Sieh, W. (2016). REVEL: An ensemble method for predicting
 200 the pathogenicity of rare missense variants. *Am. J. Hum. Genet.*, 99(4), 877–885.
 201 <https://doi.org/10.1016/j.ajhg.2016.08.016>
- 202 Jaganathan, K., Kyriazopoulou Panagiotopoulou, S., McRae, J. F., Darbandi, S. F., Knowles,
 203 D., Li, Y. I., Kosmicki, J. A., Arbelaez, J., Cui, W., Schwartz, G. B., Chow, E. D.,
 204 Kanterakis, E., Gao, H., Kia, A., Batzoglou, S., Sanders, S. J., & Farh, K. K.-H. (2019).
 205 Predicting splicing from primary sequence with deep learning. *Cell*, 176(3), 535–548.e24.
 206 <https://doi.org/10.1016/j.cell.2018.12.015>
- 207 Köhler, S., Gargano, M., Matentzoglou, N., Carmody, L. C., Lewis-Smith, D., Vasilevsky, N.
 208 A., Danis, D., Balagura, G., Baynam, G., Brower, A. M., Callahan, T. J., Chute, C. G.,
 209 Est, J. L., Galer, P. D., Ganesan, S., Griese, M., Haimel, M., Pazmandi, J., Hanauer, M.,
 210 ... Robinson, P. N. (2021). The human phenotype ontology in 2021. *Nucleic Acids Res.*,
 211 49(D1), D1207–D1217. <https://doi.org/10.1093/nar/gkaa1043>
- 212 Landrum, M. J., & Kattman, B. L. (2018). ClinVar at five years: Delivering on the promise.
 213 *Hum. Mutat.*, 39(11), 1623–1630. <https://doi.org/10.1002/humu.23641>
- 214 Lu, W., Gauthier, L. D., Poterba, T., Giacomuzzi, E., Goodrich, J. K., Stevens, C. R., King,
 215 D., Daly, M. J., Neale, B. M., & Karczewski, K. J. (2023). CHARR efficiently estimates
 216 contamination from DNA sequencing data. *Am. J. Hum. Genet.*, 110(12), 2068–2076.

- 217 <https://doi.org/10.1016/j.ajhg.2023.10.011>
- 218 Maassen, W. T. K., Johansson, L. F., Charbon, B., Hendriksen, D., Hoek, S. van den, Slofstra,
219 M. K., Mulder, R., Meems-Veldhuis, M. T., Sietsma, R., Lemmink, H. H., Diemen, C. C.
220 van, Gijn, M. E. van, Swertz, M. A., & Velde, K. J. van der. (2025). MOLGENIS VIP: An
221 end-to-end DNA variant interpretation pipeline for research and diagnostics configurable to
222 support rapid implementation of new methods. *NAR Genom. Bioinform.*, 7(2), lqaf087.
223 <https://doi.org/10.1093/nargab/lqaf087>
- 224 Martin, A. R., Williams, E., Foulger, R. E., Leigh, S., Daugherty, L. C., Niblock, O., Leong, I.
225 U. S., Smith, K. R., Gerasimenko, O., Haraldsdottir, E., Thomas, E., Scott, R. H., Baple,
226 E., Tucci, A., Brittain, H., Burca, A. de, Ibañez, K., Kasperaviciute, D., Smedley, D., ...
227 McDonagh, E. M. (2019). PanelApp crowdsources expert knowledge to establish consensus
228 diagnostic gene panels. *Nat. Genet.*, 51(11), 1560–1565. [https://doi.org/10.1038/s41588-](https://doi.org/10.1038/s41588-019-0528-2)
229 [019-0528-2](https://doi.org/10.1038/s41588-019-0528-2)
- 230 McLaren, W., Gil, L., Hunt, S. E., Riat, H. S., Ritchie, G. R. S., Thormann, A., Flicek, P., &
231 Cunningham, F. (2016). The ensembl variant effect predictor. *Genome Biol.*, 17(1), 122.
232 <https://doi.org/10.1186/s13059-016-0974-4>
- 233 Munro, J. E., Dunwoodie, S. L., & Giannoulatou, E. (2017). SVPV: A structural variant
234 prediction viewer for paired-end sequencing datasets. *Bioinformatics*, 33(13), 2032–2033.
235 <https://doi.org/10.1093/bioinformatics/btx117>
- 236 Munro, J. E., Thiagarajah, H., Bennett, M. F., Chiu, A. T. G., Schneider, A. L., Bennett, C.
237 A., Liewering, N., Allan, T., Witkowski, T., Harris, R. V., Reid, J., Sikta, N., Macdonald,
238 S., Coulter, L., Dang, Y. L., Kerkhof, J., Sadikovic, B., Perucca, P., Berkovic, S. F., ...
239 Hildebrand, M. S. (2026). Genome sequencing boosts diagnostic yield for the developmental
240 and epileptic encephalopathies. *medRxiv*, 2026.04.24.26351703. [https://doi.org/10.64898/](https://doi.org/10.64898/2026.04.24.26351703)
241 [2026.04.24.26351703](https://doi.org/10.64898/2026.04.24.26351703)
- 242 Nicholas, T. J., Cormier, M. J., & Quinlan, A. R. (2022). Annotation of structural variants
243 with reported allele frequencies and related metrics from multiple datasets using SVAfotate.
244 *BMC Bioinformatics*, 23(1), 490. <https://doi.org/10.1186/s12859-022-05008-y>
- 245 Oliver, K. L., Scheffer, I. E., Bennett, M. F., Grinton, B. E., Bahlo, M., & Berkovic, S.
246 F. (2023). Genes4Epilepsy: An epilepsy gene resource. *Epilepsia*, 64(5), 1368–1375.
247 <https://doi.org/10.1111/epi.17547>
- 248 Pais, L. S., Snow, H., Weisburd, B., Zhang, S., Baxter, S. M., DiTroia, S., O'Heir, E., England,
249 E., Chao, K. R., Lemire, G., Osei-Owusu, I., VanNoy, G. E., Wilson, M., Nguyen, K.,
250 Arachchi, H., Phu, W., Solomonson, M., Mano, S., O'Leary, M., ... O'Donnell-Luria, A.
251 (2022). Seqr: A web-based analysis and collaboration tool for rare disease genomics. *Hum.*
252 *Mutat.*, 43(6), 698–707. <https://doi.org/10.1002/humu.24366>
- 253 Pedersen, B. S., Bhetariya, P. J., Brown, J., Kravitz, S. N., Marth, G., Jensen, R. L., Bronner,
254 M. P., Underhill, H. R., & Quinlan, A. R. (2020). Somalier: Rapid relatedness estimation
255 for cancer and germline studies using efficient genome sketches. *Genome Med.*, 12(1), 62.
256 <https://doi.org/10.1186/s13073-020-00761-2>
- 257 Pedersen, B. S., Brown, J. M., Dashnow, H., Wallace, A. D., Velinder, M., Tristani-Firouzi,
258 M., Schiffman, J. D., Tvrdik, T., Mao, R., Best, D. H., Bayrak-Toydemir, P., & Quinlan, A.
259 R. (2021). Effective variant filtering and expected candidate variant yield in studies of rare
260 human disease. *NPJ Genom Med*, 6(1), 60. <https://doi.org/10.1038/s41525-021-00227-3>
- 261 Pedersen, B. S., Layer, R. M., & Quinlan, A. R. (2016). Vcfanno: Fast, flexible annotation of
262 genetic variants. *Genome Biol.*, 17(1), 118. <https://doi.org/10.1186/s13059-016-0973-5>
- 263 Rafehi, H., Read, J., Szmulewicz, D. J., Davies, K. C., Snell, P., Fearnley, L. G., Scott,
264 L., Thomsen, M., Gillies, G., Pope, K., Bennett, M. F., Munro, J. E., Ngo, K. J.,
265 Chen, L., Wallis, M. J., Butler, E. G., Kumar, K. R., Wu, K. H., Tomlinson, S. E., ...

- 266 Lockhart, P. J. (2023). An intronic GAA repeat expansion in FGF14 causes the autosomal-
267 dominant adult-onset ataxia SCA50/ATX-FGF14. *Am. J. Hum. Genet.*, 110(1), 105–119.
268 <https://doi.org/10.1016/j.ajhg.2022.11.015>
- 269 Rafehi, H., Szmulewicz, D. J., Bennett, M. F., Sobreira, N. L. M., Pope, K., Smith, K. R.,
270 Gillies, G., Diakumis, P., Dolzhenko, E., Eberle, M. A., Barcina, M. G., Breen, D. P.,
271 Chancellor, A. M., Cremer, P. D., Delatycki, M. B., Fogel, B. L., Hackett, A., Halmagyi,
272 G. M., Kapetanovic, S., ... Lockhart, P. J. (2019). Bioinformatics-based identification of
273 expanded repeats: A non-reference intronic pentamer expansion in RFC1 causes CANVAS.
274 *Am. J. Hum. Genet.*, 105(1), 151–165. <https://doi.org/10.1016/j.ajhg.2019.05.016>
- 275 Robinson, J. T., Thorvaldsdottir, H., & Mesirov, J. P. (2026). Igv-reports: Embedding
276 interactive genomic visualizations in HTML reports to aid variant review. *Bioinformatics*,
277 42(4), btag125. <https://doi.org/10.1093/bioinformatics/btag125>
- 278 Schubach, M., Maass, T., Nazaretyan, L., Röner, S., & Kircher, M. (2024). CADD v1.7:
279 Using protein language models, regulatory CNNs and other nucleotide-level scores to
280 improve genome-wide variant predictions. *Nucleic Acids Res.*, 52(D1), D1143–D1154.
281 <https://doi.org/10.1093/nar/gkad989>
- 282 Stranneheim, H., Lagerstedt-Robinson, K., Magnusson, M., Kvarnung, M., Nilsson, D., Lesko,
283 N., Engvall, M., Anderlid, B.-M., Arnell, H., Johansson, C. B., Barbaro, M., Björck, E.,
284 Bruhn, H., Eisfeldt, J., Freyer, C., Grigelioniene, G., Gustavsson, P., Hammarsjö, A.,
285 Hellström-Pigg, M., ... Wedell, A. (2021). Integration of whole genome sequencing into
286 a healthcare setting: High diagnostic rates across multiple clinical entities in 3219 rare
287 disease patients. *Genome Med.*, 13(1), 40. <https://doi.org/10.1186/s13073-021-00855-5>
- 288 Wallis, M., Bodek, S. D., Munro, J., Rafehi, H., Bennett, M. F., Ye, Z., Schneider, A.,
289 Gardiner, F., Valente, G., Murdoch, E., Uebergang, E., Hunter, J., Stutterd, C., Huq,
290 A., Salmon, L., Scheffer, I., Eratne, D., Meyn, S., Fong, C. Y., ... Berkovic, S. F. (2024).
291 Experience of the first adult-focussed undiagnosed disease program in australia (AHA-UDP):
292 Solving rare and puzzling genetic disorders is ageless. *Orphanet J. Rare Dis.*, 19(1), 288.
293 <https://doi.org/10.1186/s13023-024-03297-5>